

Week 3 : fortigate vpn with SD-wan

what is sd - wan?

SD-WAN (Software-Defined Wide Area Network) is a modern approach to networking that allows businesses to connect their branch offices, data centers, and cloud resources using software intelligence rather than relying solely on dedicated hardware circuits

SD-WAN Purpose

SD-WAN allows your FortiGate to:

Combine multiple WAN connections

Examples:

- WAN1 = Fiber
- WAN2 = 4G
- WAN3 = DSL

FortiGate treats them as **one virtual WAN interface**.

Automatically choose the fastest / best link

FortiGate constantly checks:

- latency
- jitter
- packet loss
- bandwidth

and chooses the **best WAN link for the traffic**.

This is why SD-WAN is perfect for VPN.

Failover — If one WAN goes down, traffic moves automatically

Example:

- WAN1 fails → traffic moves to WAN2 instantly
- VPN tunnel stays up
- Users don't feel anything

Balance traffic across multiple WAN links

You can define policies:

- YouTube → WAN2
- VPN → WAN1
- Web browsing → any link

This optimizes performance and cost.

Improve VPN stability

SD-WAN can route your IPsec VPN through:

- the fastest WAN
- the WAN with lowest latency
- the path that meets SLA (Service Level Agreement)

Configuration of SD-WAN on fortigate

Figure 1:configure the ports to a wan

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Packet Capture

SD-WAN

Static Routes

Policy Routes

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Log & Report

FortiGate time is out of sync.

Edit Interface

Name

wan 2 (port3)

Alias

wan 2

Type

Physical Interface

VRF ID

0

Role

WAN

Estimated bandwidth

0

kbps Upstream

0

kbps Downstream

Address

Addressing mode

Manual

DHCP

Status

Connected

Obtained IP/Netmask

30.30.30.2/255.255.255.0

Renew

Expiry Date

2025/12/04 10:39:13

Acquired DNS

192.168.113.1

Default gateway

30.30.30.1

Retrieve default gateway from server

Distance

5

Override internal DNS

Administrative Access

IPv4

HTTPS

PING

FMG-Access

OK

Cancel

FortiGate

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Status

Up

MAC address

00:0c:29:ab:10:df

Speed Test

Execute speed test

Additional Information

API Preview

References

Edit in CLI

Documentation

Online Help

Video Tutorials

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FortiGate time is out of sync.

FortiGate VM64

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			
port8	Physical Interface		0.0.0.0/0.0.0.0			
port9	Physical Interface		0.0.0.0/0.0.0.0			
port10	Physical Interface		0.0.0.0/0.0.0.0			
WAN (port4)	Physical Interface		1.1.1.1/255.255.255.0	PING		
wan 2 (port3)	Physical Interface		30.30.30.2/255.255.255.0	PING		
SD-WAN Zone						
virtual-wan-link	SD-WAN Zone	WAN (port4) wan 2 (port3)	0.0.0.0/0.0.0.0			
Tunnel Interface						
NAT interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0			

Security Rating Issues

100% Updated: 12:15:31

Figure 2 : create new sd-wan member

The image displays two screenshots of the FortiGate SD-WAN configuration interface. The top screenshot shows the 'Edit SD-WAN Member' dialog box. The bottom screenshot shows the 'SD-WAN Zones' overview page.

Top Screenshot: Edit SD-WAN Member

The dialog box is titled 'Edit SD-WAN Member'. It contains the following fields:

- Interface: ☐ None
- SD-WAN Zone:
- Gateway:
- Cost:
- Priority:
- Status: ☒ Enabled ☐ Disabled

Additional Information on the right includes:

- API Preview
- SD-WAN Setup Guides
 - Creating the SD-WAN Interface
 - MPLS (SIP and Backup) + DIA (Cloud Apps)
 - SD-WAN Traffic Shaping and QoS with SD-WAN
 - Per Packet Distribution and Tunnel Aggregation
- Documentation
 - Online Help
 - Video Tutorials

Buttons at the bottom: OK, Cancel.

Bottom Screenshot: SD-WAN Zones

The overview page shows the 'SD-WAN Zones' tab. It displays two donut charts for 'Download' and 'Upload' bandwidth. Both charts show a total of 2, with a breakdown of port4 (green) and port3 (orange).

Interfaces	Gateway	Cost	Download	Upload
virtual-wan-link				

Buttons: + Create New, Edit, Delete.

Footer: Updated: 09:21:29

figure 3: Define Link Performance Metrics (Performance SLA)

Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.

The screenshot shows the FortiGate SD-WAN configuration interface. The left sidebar lists various configuration categories, with 'SD-WAN' highlighted. The main panel displays the 'Edit Performance SLA' configuration for a rule named 'VPN_Quality_Monitor'. The configuration includes settings for the probe mode (Active), protocol (Ping), server (1.1.1.2), and participants (All SD-WAN Members). The SLA Target section shows thresholds for Latency (100 ms), Jitter (30 ms), and Packet Loss (1 %). The Link Status section shows the check interval (500 ms), failures before inactive (5), and restore link after (5 check(s)). The Actions when Inactive section shows the 'Update static route' option enabled. The right sidebar shows the SLA Details table and additional information links.

	Packet Loss	Latency	Jitter
WAN (port4)	0.00%	1.66ms	1.63ms
wan 2 (port3)	0.00%	1.66ms	1.63ms

Additional Information

- API Preview
- Edit in CLI
- Performance SLA Setup Guides
- Link Monitoring
- SLA Targets
- Documentation
- Online Help
- Video Tutorials

Figure 4 : Create the SD-WAN Rule (Traffic Steering Logic)

Purpose: This rule specifically directs the VPN traffic to use the **best performing** link based on the real-time data collected

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FortiGate time is out of sync.

SD-WAN ZonesSD-WAN RulesPerformance SLAs

Create NewEditCloneDeleteSearch

ID	Name	Source	Destination	Criteria	Members	Hit Count
1	Optimize_Alex_VPN	NET 10	NET 30	Latency	WAN (port4)wan 2 (port3)	0
Implicit						
	sd-wan	all	all	Source IP	any	

Updated: 09:25:10

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Priority Rule

NameOptimize_Alex_VPN

Source

Source addressNET 10

User group

Destination

AddressNET 30

Protocol numberTCPUDPANYSpecify0

Internet Service

Application

Outgoing Interfaces

Select a strategy for how outgoing interfaces will be chosen.

Manual

Manually assign outgoing interfaces.

Best Quality

The interface with the best measured performance is selected.

Lowest Cost (SLA)

ID1

Last used2025/11/28 14:11:40

Hit count0

Additional Information

API Preview

Edit in CLI

SD-WAN Rules Setup Guides

Implicit Rule

Best Quality

Lowest Cost (SLA)

Maximize Bandwidth (SLA)

Documentation

Online Help

Video Tutorials

OKCancel

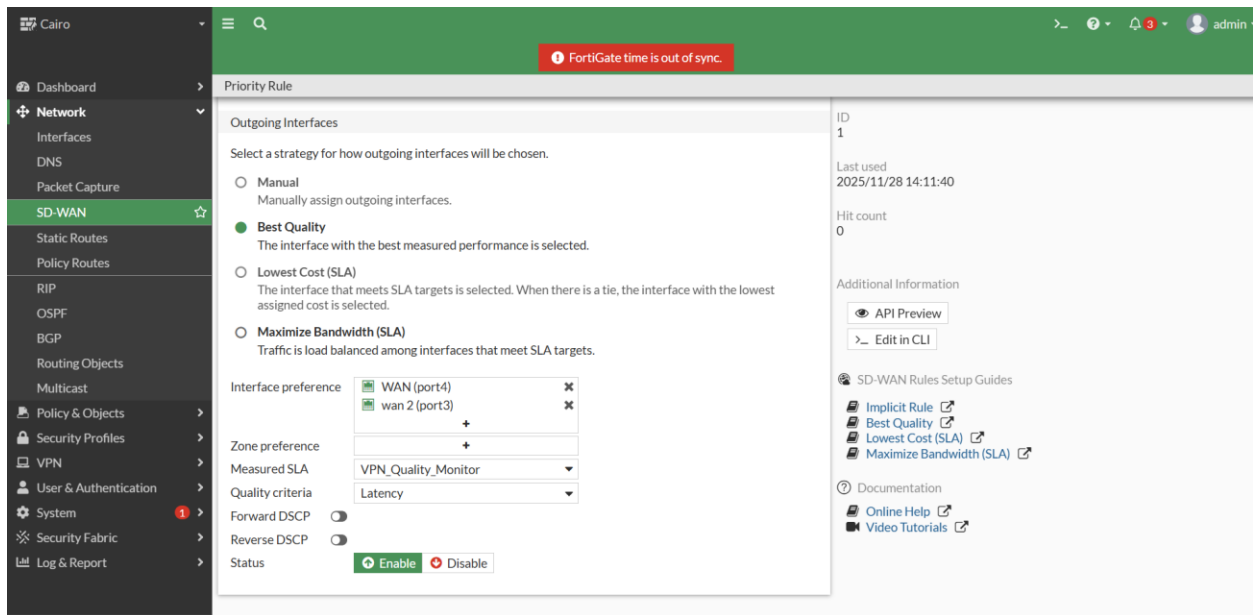
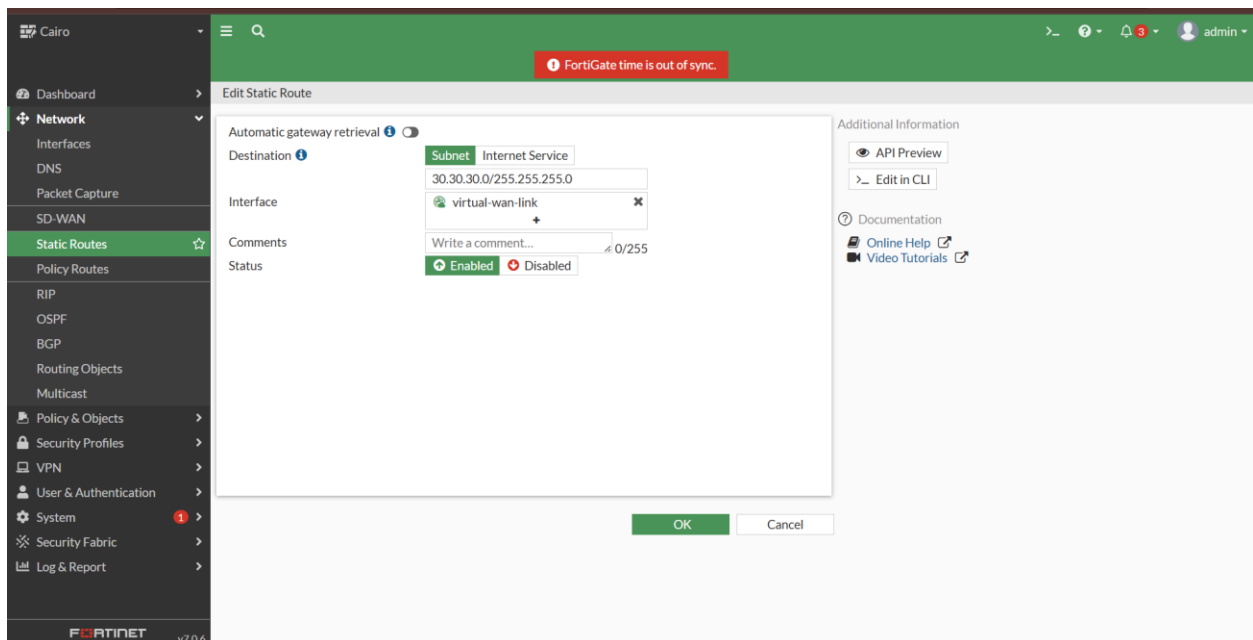


Figure 5 : Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.



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FortiGate time is out of sync.

Create NewEditCloneDeleteSearch

Destination	Gateway IP	Interface	Status	Comments
30.30.30.0/24		virtual-wan-link	Enabled	

FortiGate v7.0.6